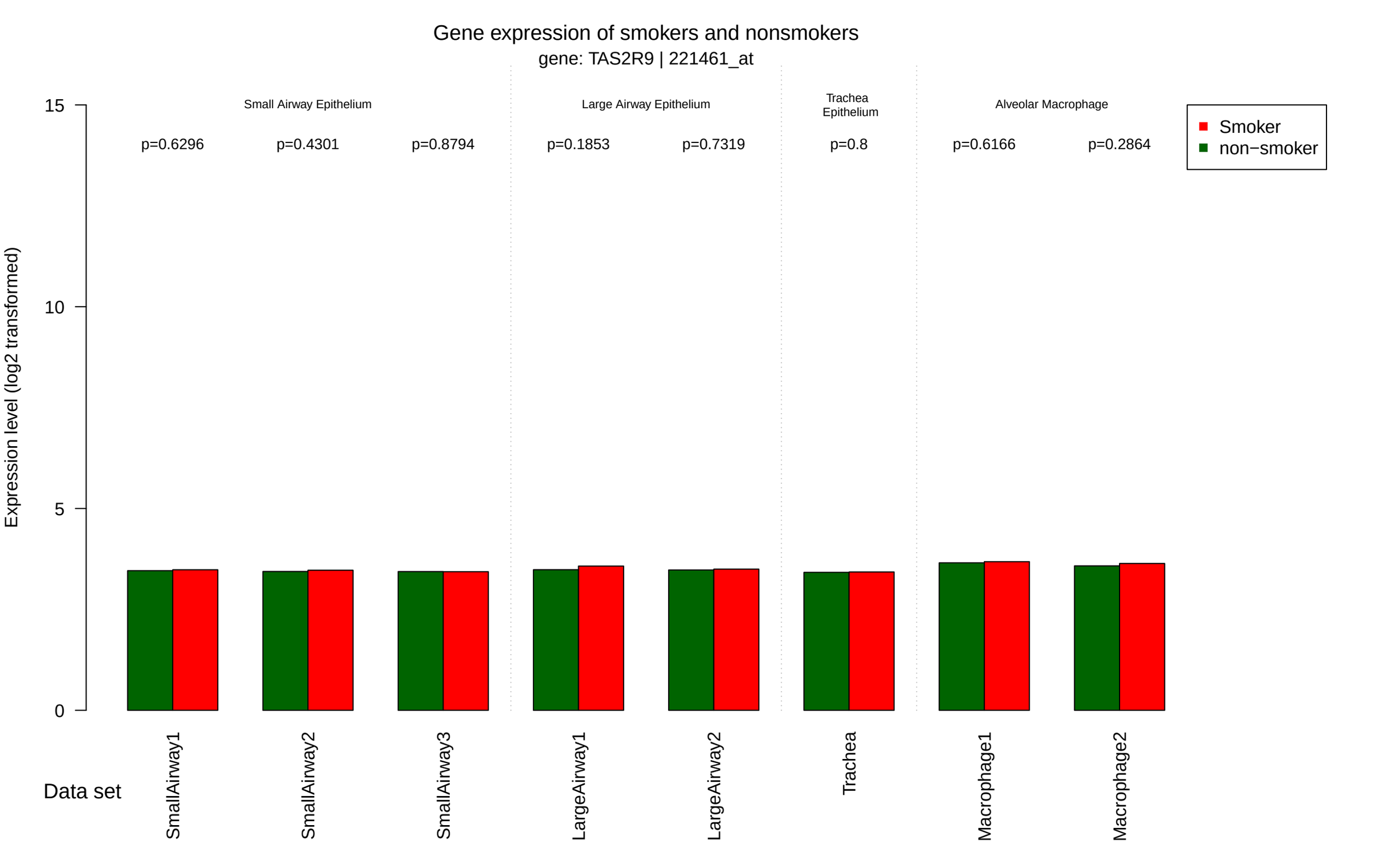
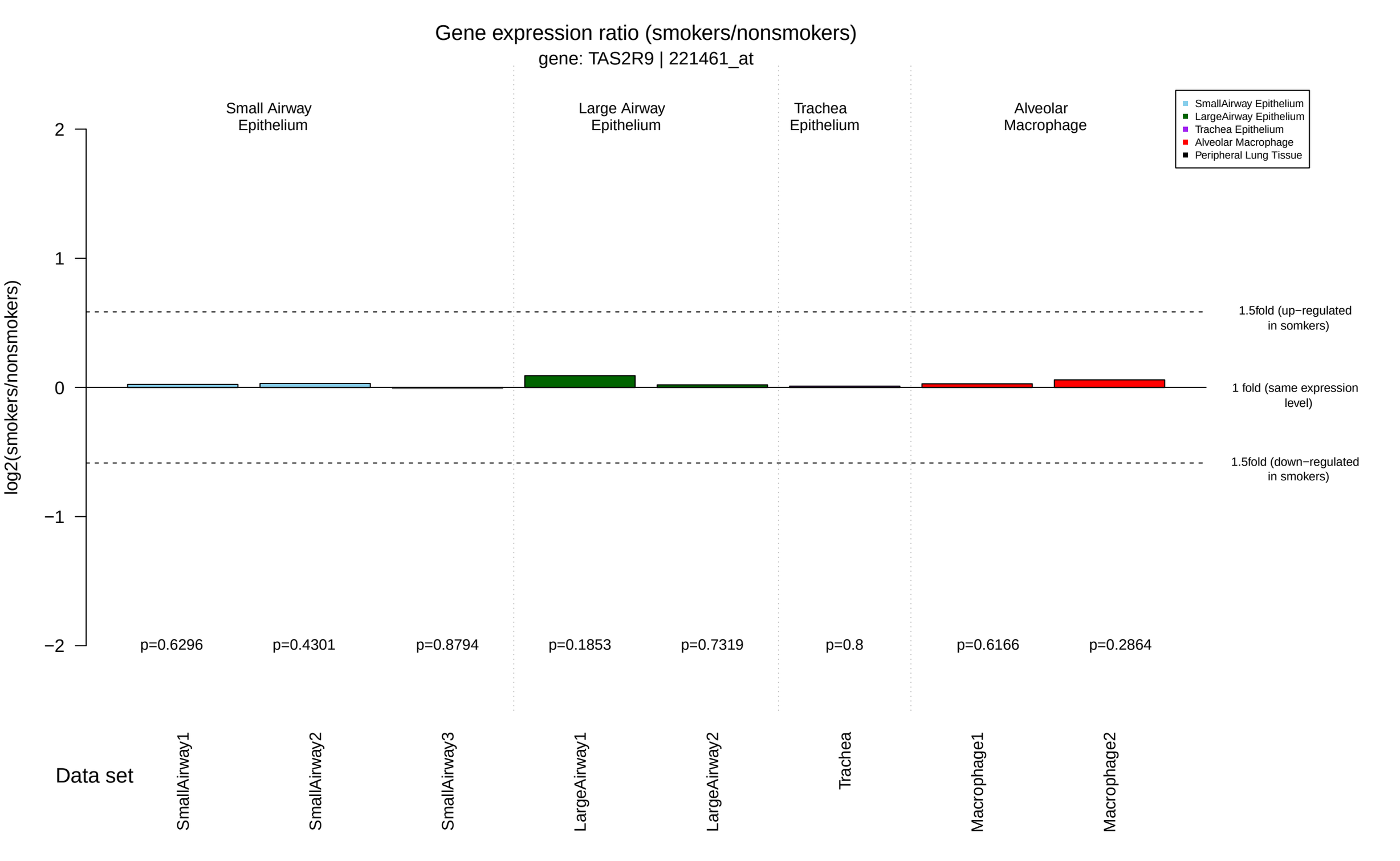
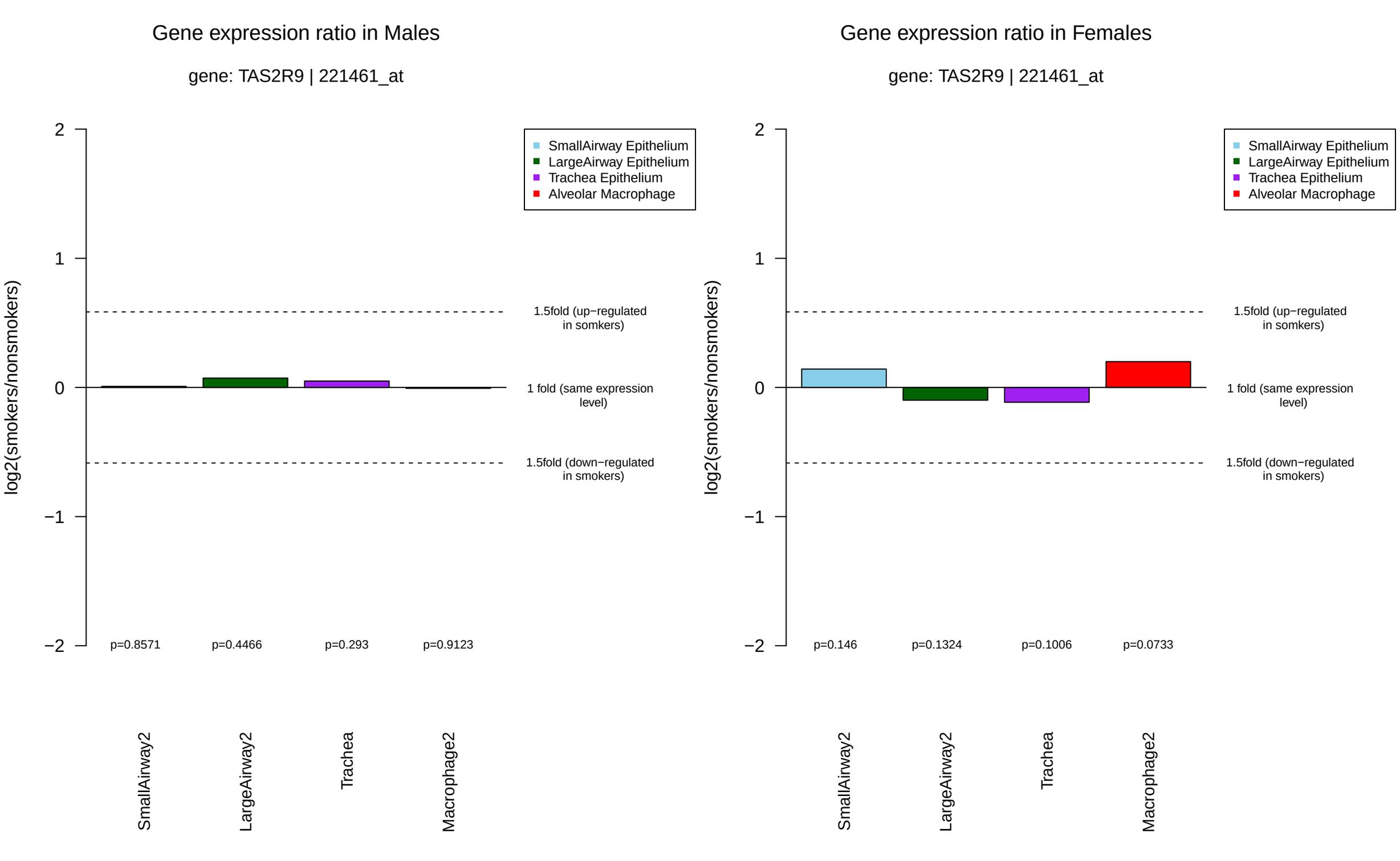
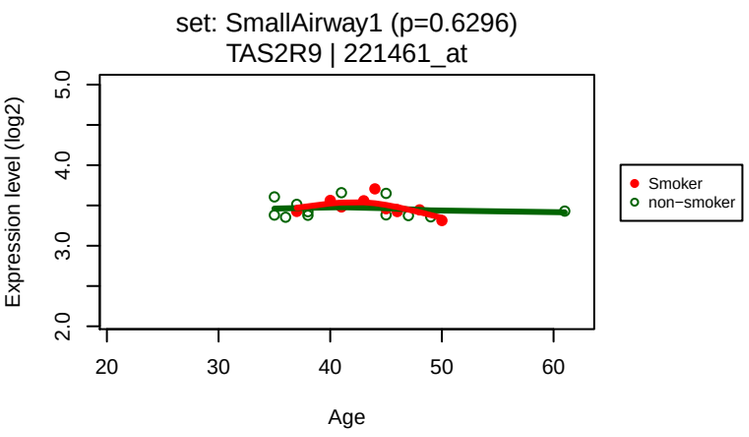
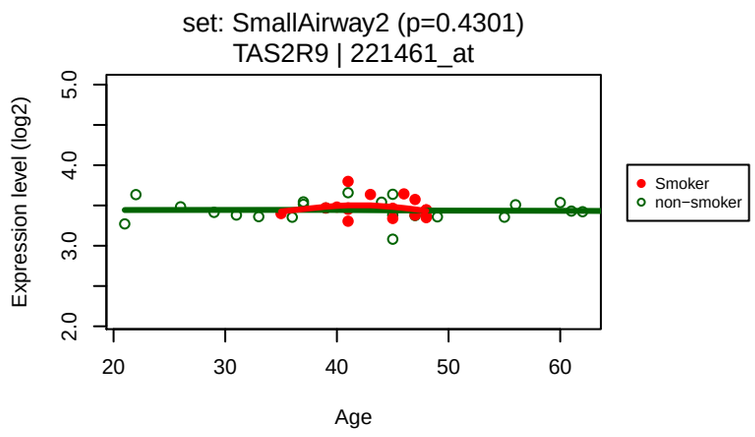
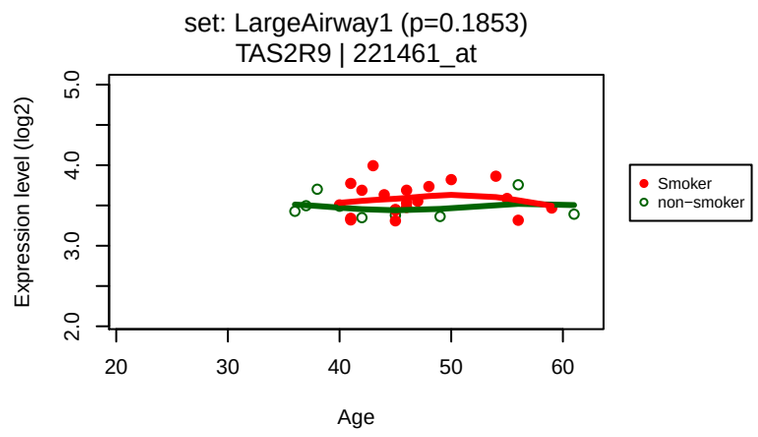

Gene expression of smokers and nonsmokers
gene: TAS2R9 | 221461_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6296
p=0.4301
p=0.8794
p=0.1853
p=0.7319
p=0.8
p=0.6166
p=0.2864
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

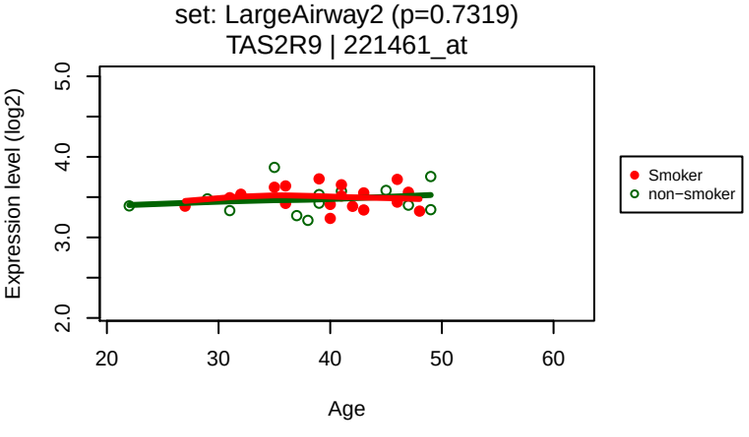

Gene expression ratio (smokers/nonsmokers)
gene: TAS2R9 | 221461_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6296
p=0.4301
p=0.8794
p=0.1853
p=0.7319
p=0.8
p=0.6166
p=0.2864
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

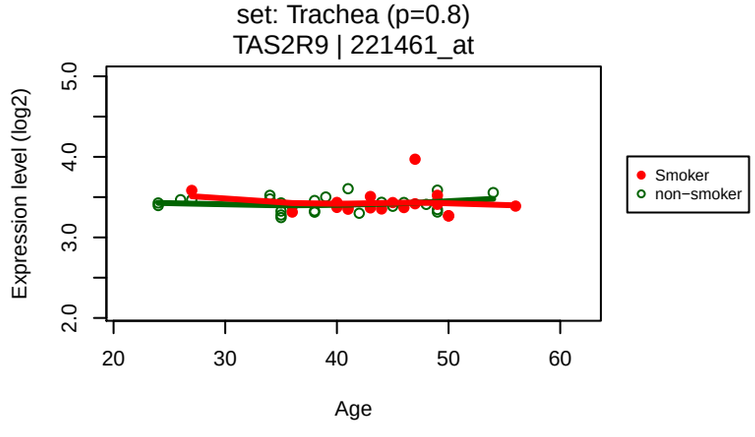

Gene expression ratio in Males
gene: TAS2R9 | 221461_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8571
p=0.4466
p=0.293
p=0.9123
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS2R9 | 221461_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.146
p=0.1324
p=0.1006
p=0.0733
SmallAirway2
LargeAirway2
Trachea
Macrophage2

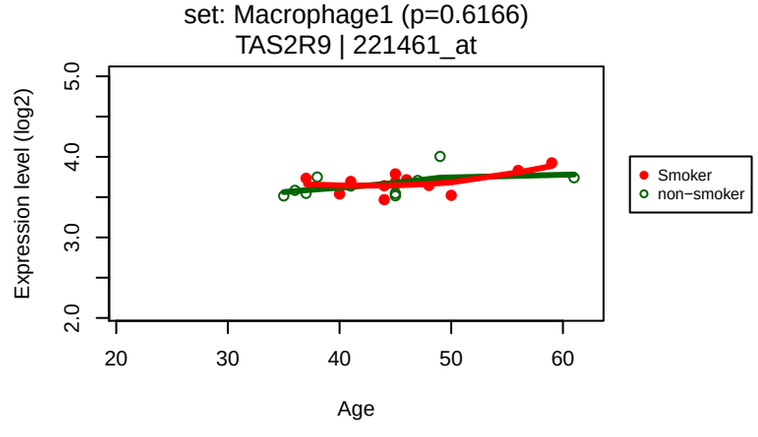

set: SmallAirway1 (p=0.6296)
TAS2R9 | 221461_at
Expression level (log2)
Age
Smoker
non−smoker

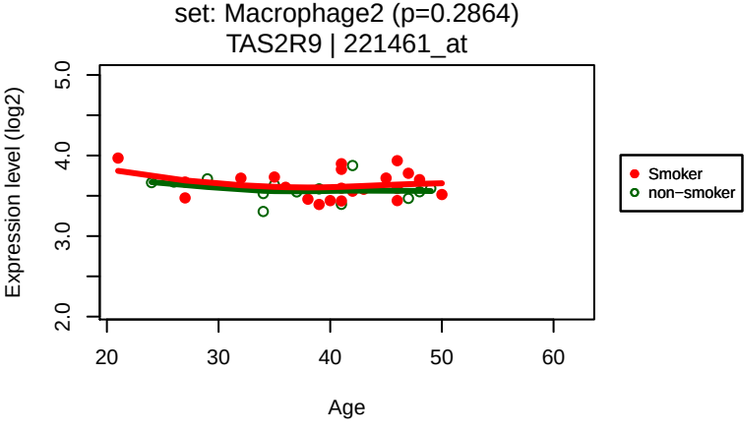

set: SmallAirway2 (p=0.4301)
TAS2R9 | 221461_at
Expression level (log2)
Age
Smoker
non−smoker

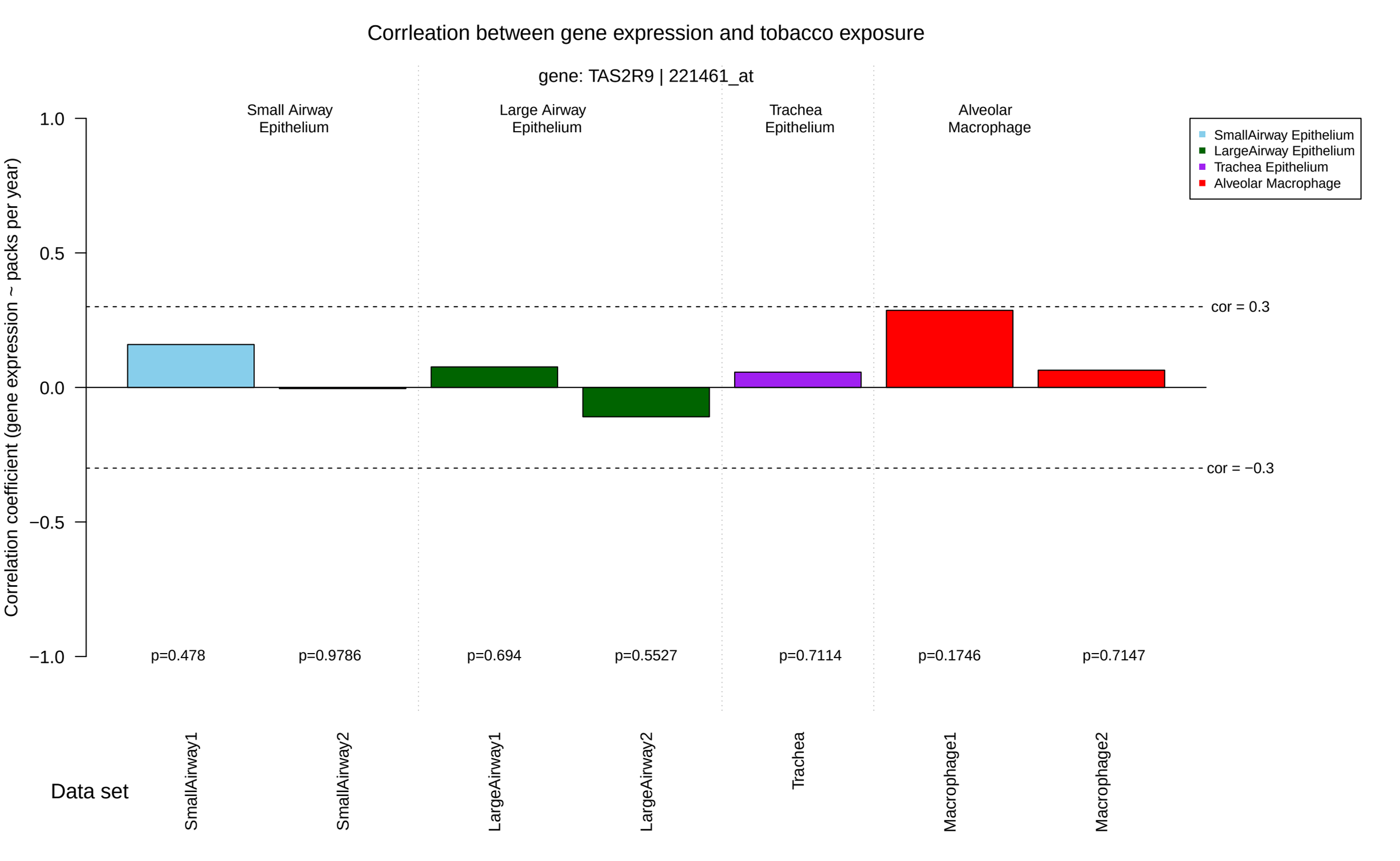

set: LargeAirway1 (p=0.1853)
TAS2R9 | 221461_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7319)
TAS2R9 | 221461_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8)
TAS2R9 | 221461_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6166)
TAS2R9 | 221461_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2864)
TAS2R9 | 221461_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS2R9 | 221461_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.478
p=0.9786
p=0.694
p=0.5527
p=0.7114
p=0.1746
p=0.7147
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2